use your Google account to sign up for an account at The Weather Channel. When you authenticate, you're redirected to Google, where you log in and Google returns a token with information about you, called *claims* in the world of identity, to the Weather Channel.

There are several advantages to using federated identity. First, the RP doesn't have to maintain authentication systems or support onboarding. Second, people find it more convenient to just sign in with an account that already exists instead of creating a new one. Finally, because these protocols are widespread, they are well supported by identity systems and there are many development libraries to choose from.

The risks of federated identity primarily revolve around the dependence on an external party for a critical function. This is not as big a risk in enterprise federated use cases like the benefits provider we discussed earlier, but can be a significant risk for RPs in online use cases. Consequently, most IdPs are big organizations like Google, Amazon, Facebook, and Apple that the RP can trust to be reliable. The primary risk for principals is deplatforming: If you lose your Google account, for example, you also lose access to all the places you've used it to sign up.

Another significant risk is the loss of privacy when a federated identity provider (IdP) can see every site or service a person accesses. This kind of cross-service surveillance may not be as overtly disruptive as deplatforming, but it creates a persistent record of a person's online activity. Over time, that undermines trust—people may begin to hesitate before engaging online, knowing that their interactions are being silently tracked. Personally, I always use the email option of federated identity whenever I can to avoid both of these risks.

The RP's lack of control over the identity attributes that the IdP shares poses a risk to authorization systems, where access to the right attributes is often crucial to making good decisions. This requires authorization systems that rely on decentralized identity models to have the means of augmenting the data they receive from the IdP with other needed attributes.